Chapter 19



Spontaneous

Nonspontaneous



Chemical Thermodynamics

- 19.1 Spontaneous Processes
- 19.2 Entropy and the Second Law of Thermodynamics
- 19.3 Molecular Interpretation of Entropy and the Third Law
- 19.4 Entropy Changes in Chemical Reactions
- 19.5 Gibbs Free Energy
- 19.6 Free Energy and Temperature
- 19.7 Free Energy and the Equilibrium Constant

19.1 Spontaneous Processes

Enthalpy (H) and First law of thermodynamics (Review)

- Enthalpy is the internal energy of a system plus the product of its pressure and volume

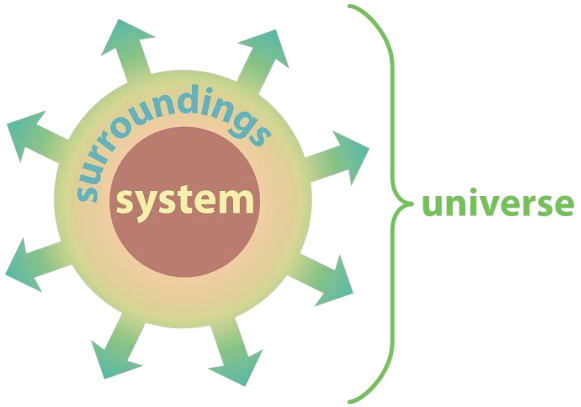
$$H = E + PV$$

Endothermic reaction: $\Delta H > 0$

ΔH is the energy absorbed by the reaction

Exothermic reaction: $\Delta H < 0$

ΔH is the energy given off by the reaction



First law of thermodynamics

- Energy lost by the system = energy gained by the surroundings
- Energy can be converted from one form to another
- Energy is conserved, but that alone does not tell us which way a change will go.

19.1 Spontaneous Processes

Spontaneous Processes

Q. An egg falls and breaks. Will the broken egg ever reassemble itself and jump back into your hand?



Spontaneous

Nonspontaneous



- A **spontaneous process** (thermodynamically favored) proceeds on its own, without continuous outside assistance.
- A process that is spontaneous in one direction is **nonspontaneous** in the reverse direction (at the same conditions!).

- Example: an iron nail rusts on its own; rust never turns back into a shiny nail.

✓ **Spontaneous ≠ fast**: some spontaneous reactions are very slow (diamond → graphite; rusting).

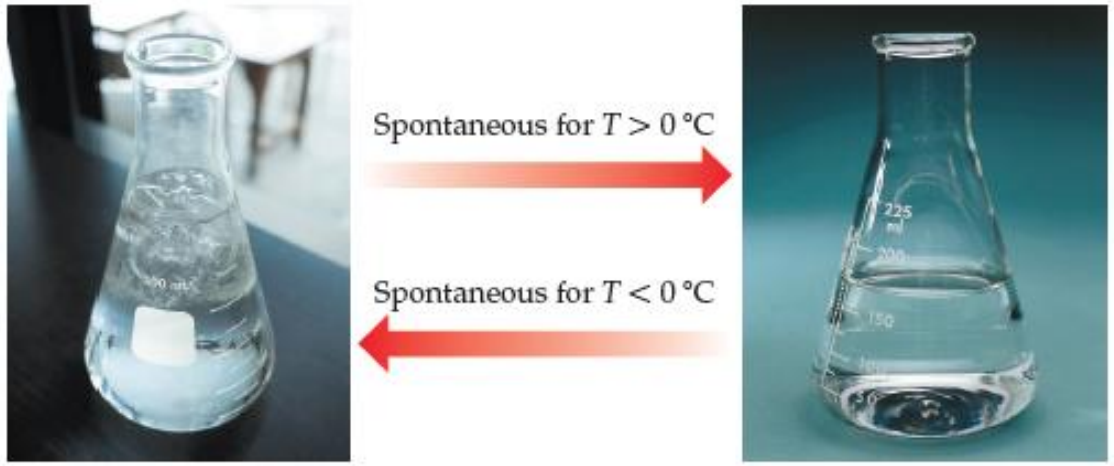
✓ **Nonspontaneous ≠ impossible**: it can be driven by a continuous input of energy (e.g. electrolysis of water).

19.1 Spontaneous Processes

Temperature Affects Spontaneity

Q. Is the melting of ice spontaneous?

- Experimental conditions (T, P) can change the direction of spontaneous change.



- It depends on temperature!
- Above 0 °C: ice melts spontaneously.
- Below 0 °C: water freezes spontaneously.
- At 0 °C (1 atm): ice and water are at **equilibrium** – neither direction is favoured.

- The boiling point of water also changes with pressure. How?
- Thermodynamics tells us the **direction and extent of a reaction**, not its speed.

19.1 Spontaneous Processes

Q. Predict whether each process is spontaneous as described, spontaneous in the reverse direction, or at equilibrium:

(a) Water at 40 °C gets hotter when a piece of metal heated to 150 °C is added.

(b) Water at room temperature decomposes into $H_2(g)$ and $O_2(g)$.

(c) Benzene vapour at 1 atm condenses to liquid benzene at its normal boiling point, 80.1 °C.

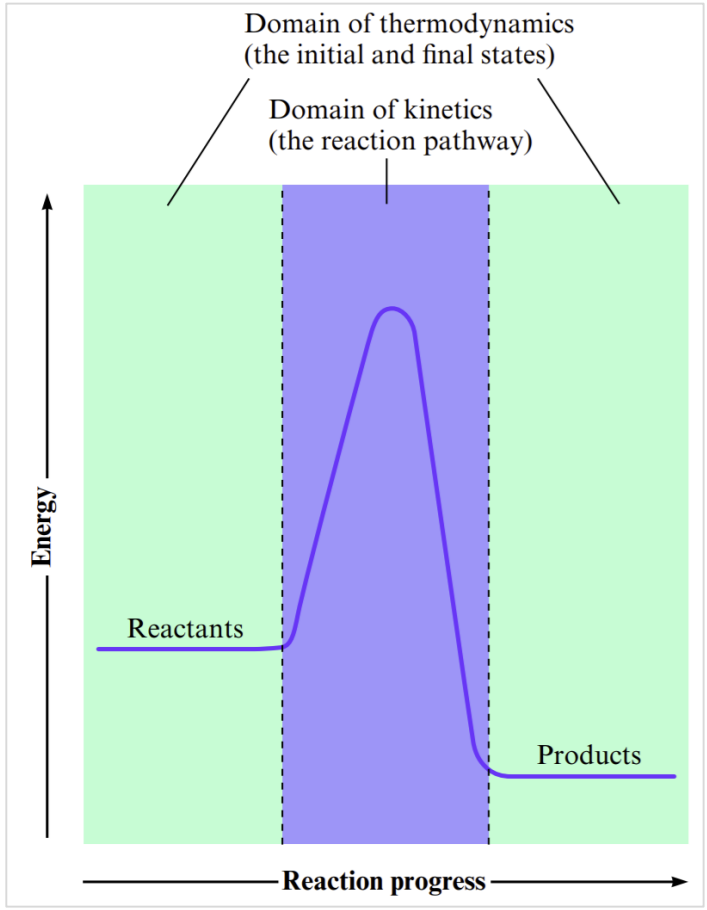
(a) **Spontaneous**: heat flows from the hotter metal to the cooler water.

(b) **Nonspontaneous**: the reverse reaction ($H_2 + O_2 \rightarrow H_2O$) is spontaneous.

(c) **Equilibrium**: at the normal boiling point, liquid and vapour coexist at 1 atm.

19.1 Spontaneous Processes

Seeking a Criterion for Spontaneity: Is the First Law Enough?



(Source: Chemistry: Steven S. Zumdahl, Susan A. Zumdahl, and Donald J. DeCoste)

- Most spontaneous processes are exothermic.

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$, $\Delta H^o = -890.3$ kJ/mol

$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$, $\Delta H^o = -824.2$ kJ/mol

Q. Does this apply to all reactions?

$H_2O(\mathbf{s}) \rightarrow H_2O(\mathbf{l})$, $\Delta H^o = 6.01$ kJ/mol

$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq)$, $\Delta H^o = 25.39$ kJ/mol

- ΔH alone cannot predict spontaneity, **entropy** is needed.
- **Entropy, S:** the tendency of energy and matter to become more disordered (spread out).

- **First law**: ΔE = q + w (tells us energy is conserved, but **not** whether a process will happen).

19.1 Spontaneous Processes

How the idea of entropy arose

1824

Sadi Carnot

- Analyzed what limits the efficiency of steam engines.
- The energy in fuel can never be converted completely to work.
- Some heat is **always lost** to the surroundings.

q/T

Rudolf Clausius

- Extended Carnot's work and found a special relationship.
- The ratio of the heat delivered to an ideal engine to the temperature at which it is delivered, q/T.
- He named this ratio **entropy**.

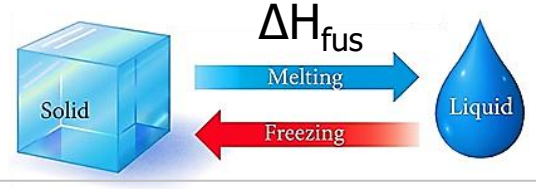
19.1 Spontaneous Processes

Reversible and Irreversible Processes

Q. Can every process be undone by exactly reversing it?

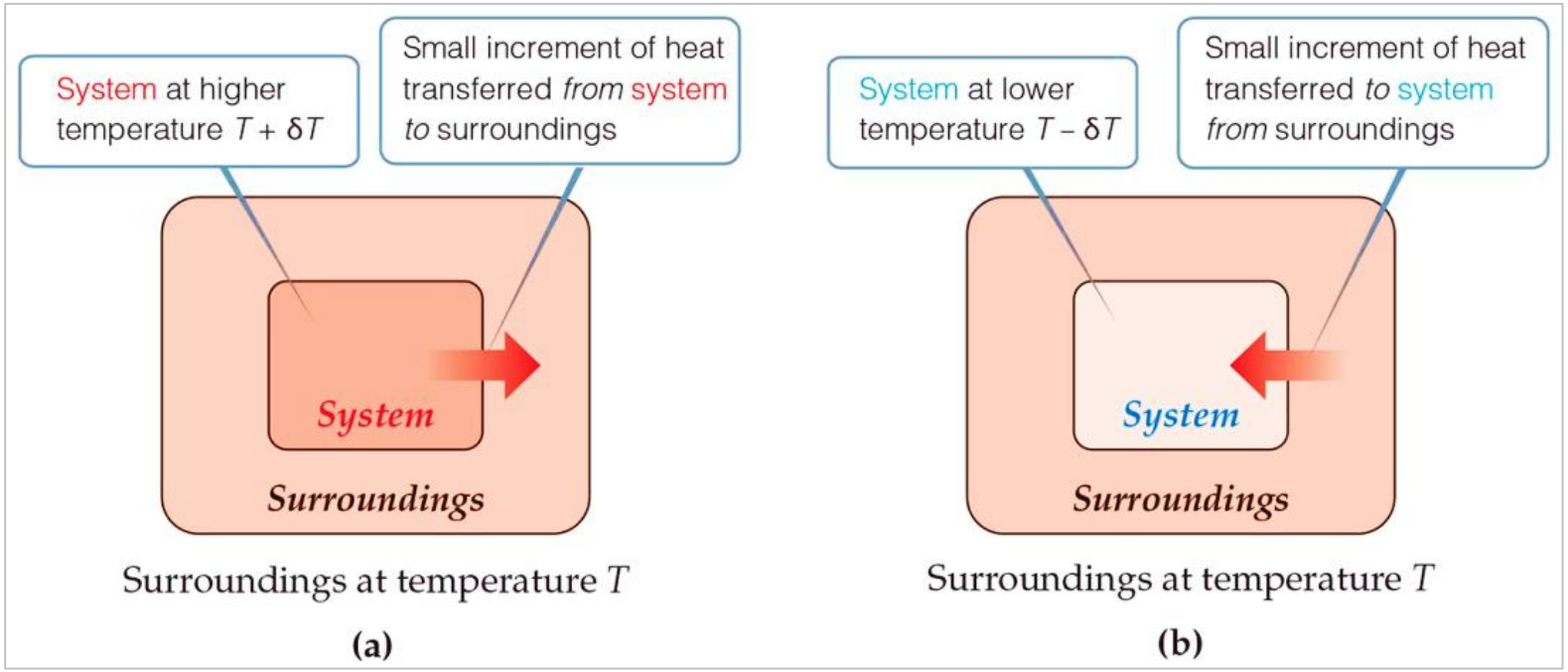
Reversible process

- System and surroundings can be restored to their original states by reversing the change.



Note:

- T, E and H are state functions.
- q and w are **not** state functions – their values depend on the path taken (reversible or irreversible).



Reversible flow of heat.

(a) Increasing the system's temperature transfers heat to the cooler surroundings.

(b) Decreasing the system's temperature transfers heat from the warmer surroundings to the system.

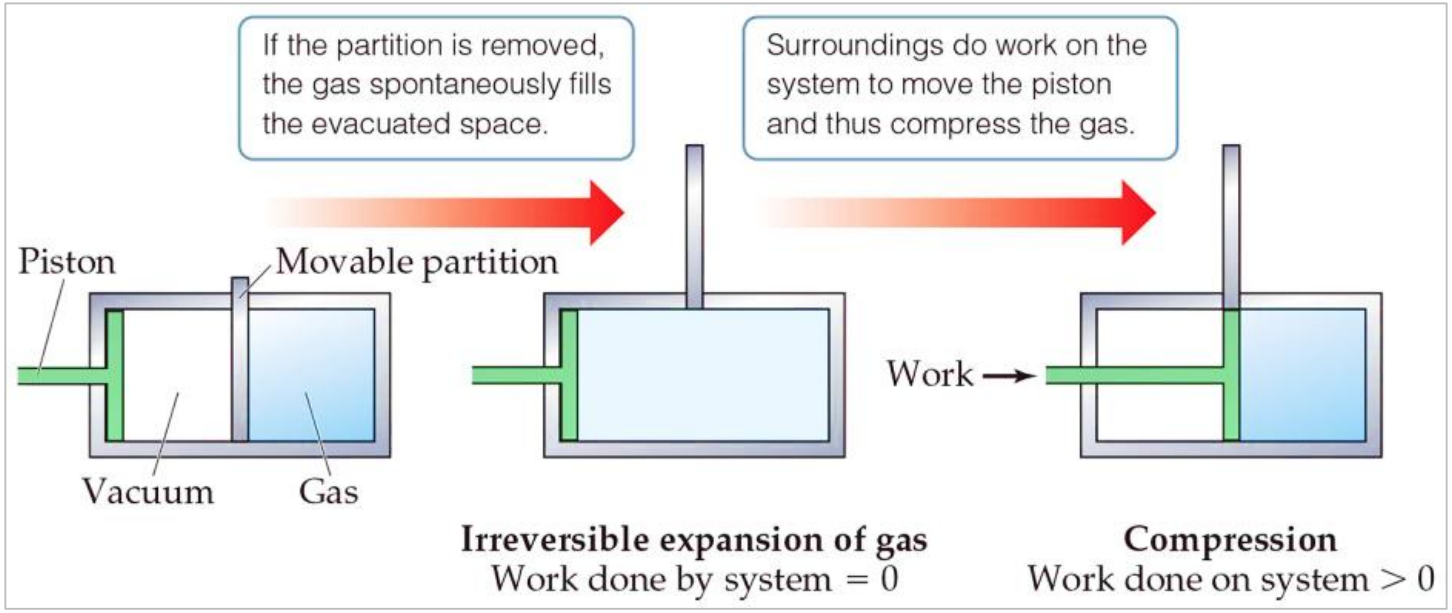
19.1 Spontaneous Processes

Reversible and Irreversible Processes

Q. Can every process be undone by exactly reversing it?

Irreversible process

- Cannot simply be reversed to restore the original states.
- All spontaneous processes are irreversible.



Note:

- T, E and H are state functions.
- q and w are **not** state functions – their values depend on the path taken (reversible or irreversible).

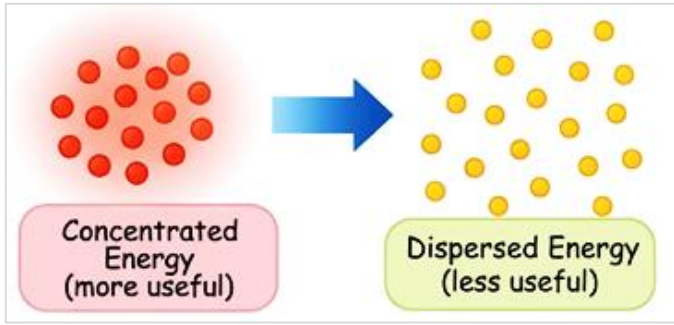
- ✓ When the partition is removed, the gas spontaneously expands to fill the whole cylinder.
- ✓ Using the piston to compress the gas back to its original state requires the surroundings to do work on the system.
- ✓ **Even if the system is returned to its original state, the surroundings have changed.**

19.1 Spontaneous Processes

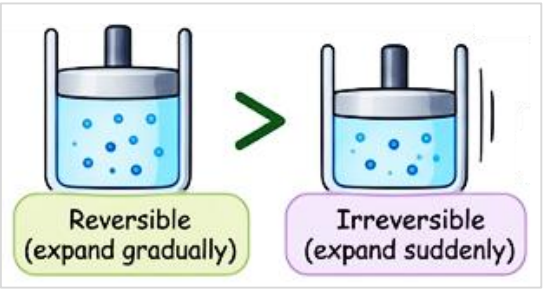
Reversible and Irreversible Processes: Maximum Work

Energy dispersion and work

- Energy spreads **spontaneously** from a concentrated to a more dispersed form (e.g. hot tea cools until it reaches the temperature of its surroundings).
- As energy disperses, less of it is available to do work.
- All real processes are irreversible.
- A reversible process produces the **maximum work** a system can do on its surroundings.





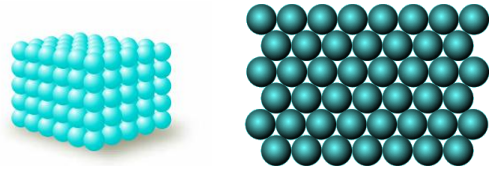


19.2 Entropy and the Second Law of Thermodynamics

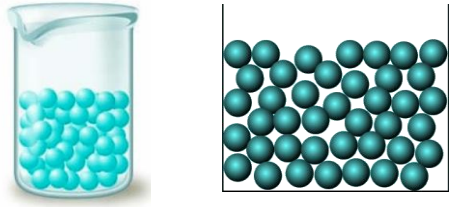
Q. How can we measure a change in entropy?

- Entropy is the degree of randomness or disorder of a system.

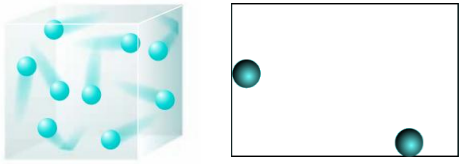
Solid



Liquid



Gas



- **Entropy (S)** is associated with how widely energy and matter are dispersed: the more ways they can be distributed, the higher S.
- The entropy of a system is a **state function** (units J/K).

$$\Delta S = S_{final} - S_{initial}$$

- If $\Delta S > 0$ the randomness increases, and if $\Delta S < 0$ the order increases.
- Increase in the entropy of the universe is the driving force for **a spontaneous process**.

19.2 Entropy and the Second Law of Thermodynamics

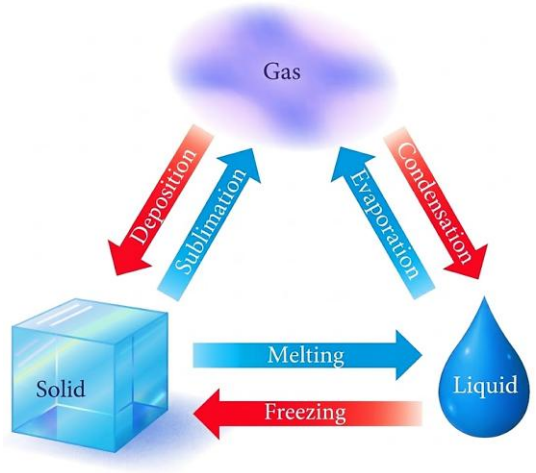
The Relationship between Entropy and Heat

- For a process at constant temperature (isothermal), ΔS equals the heat transferred **reversibly** divided by the absolute temperature (T):

$$\Delta S = \frac{q_{rev}}{T} \qquad (\text{constant } T)$$

ΔS for Phase Changes

- **Phase changes** occur reversibly at constant T (at the melting or boiling point):



$$\Delta S_{fus} = \Delta H_{fus} / T_m$$
$$\Delta S_{vap} = \Delta H_{vap} / T_b$$

Q. If the same amount of heat is transferred reversibly to water, does the entropy increase more at a lower temperature or at a higher temperature?
Compare the entropy when 100 kJ of heat is reversibly transferred to water set at 0 °C and at 100 °C.

- The same amount of heat produces a larger entropy increase at a lower temperature

19.2 Entropy and the Second Law of Thermodynamics

ΔS for Phase Changes

Q. Mercury's normal freezing point is −38.9 °C and its molar enthalpy of fusion is ΔH_{fus} = 2.29 kJ/mol. What is the entropy change of the system when 50.0 g of Hg(l) freezes at the normal freezing point?

Step 1. Moles of Hg: 50.0 g × (1 mol / 200.59 g) = 0.249 mol

Step 2. Freezing releases heat: $q_{rev} = -(0.249\ mol)(2.29 \times 10^3\ J/mol) = -571\ J$

Step 3. T = −38.9 + 273.15 = 234.3 K

Step 4. $\Delta S_{sys} = q_{rev}/T$ = −571 J / 234.3 K = **−2.44 J/K** (negative: the liquid becomes an ordered solid)

Q. Ethanol, C_2H_5OH, boils at 78.3 °C with ΔH_{vap} = 38.56 kJ/mol. Find ΔS for the system when 68.3 g of $C_2H_5OH(g)$ at 1 atm condenses to liquid at the normal boiling point (*Ans.* −163 J/K).

19.2 Entropy and the Second Law of Thermodynamics

The Second Law of Thermodynamics

Q. What determines whether a process is spontaneous?

Second law of thermodynamics:

- A reversible process produces no overall change in the entropy of the universe.
- The entropy of the universe **increases** in any spontaneous process.
- The entropy of an isolated system tends to increase.

Reversible process	**Irreversible (spontaneous) process**
$\Delta S_{univ} = \Delta S_{system} + \Delta S_{surroundings} = 0$	$\Delta S_{univ} = \Delta S_{system} + \Delta S_{surroundings} > 0$

- $\mathbf{\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}}$
- ΔS_{sys} can be negative in a spontaneous process (provided ΔS_{surr} is more positive).
- $\Delta S_{univ} < 0$ is impossible (the process is spontaneous in the reverse direction).

19.2 Entropy and the Second Law of Thermodynamics

The Second Law of Thermodynamics

Q. One mole of ice (ΔH_{fus} = 6.01 kJ/mol) melts at 0 °C (273 K) in the palm of your hand, which is at body temperature (37 °C = 310 K). Calculate ΔS_{sys}, ΔS_{surr} and ΔS_{univ}. Is the process spontaneous?

Step 1. $\Delta S_{sys} = q_{rev}/T$ = (1 mol)(6.01 × 10^3 J/mol)/273 K = **+22.0 J/K**

Step 2. The palm loses the same heat: ΔS_{surr} = (−6.01 × 10^3 J)/310 K = **−19.4 J/K**



Step 3. $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$ = 22.0 + (−19.4) = **+2.6 J/K** → spontaneous (irreversible)

When would the melting be reversible and ΔS_{univ} = 0?

19.2 Entropy and the Second Law of Thermodynamics

The Second Law of Thermodynamics

Q. When water freezes at –10 °C, the molecules become more ordered (the system's entropy decreases). Yet the process is spontaneous. How can that be?

> **Answer:** Freezing releases heat, which increases the entropy of the surroundings by more than the system loses – so the entropy of the universe increases.

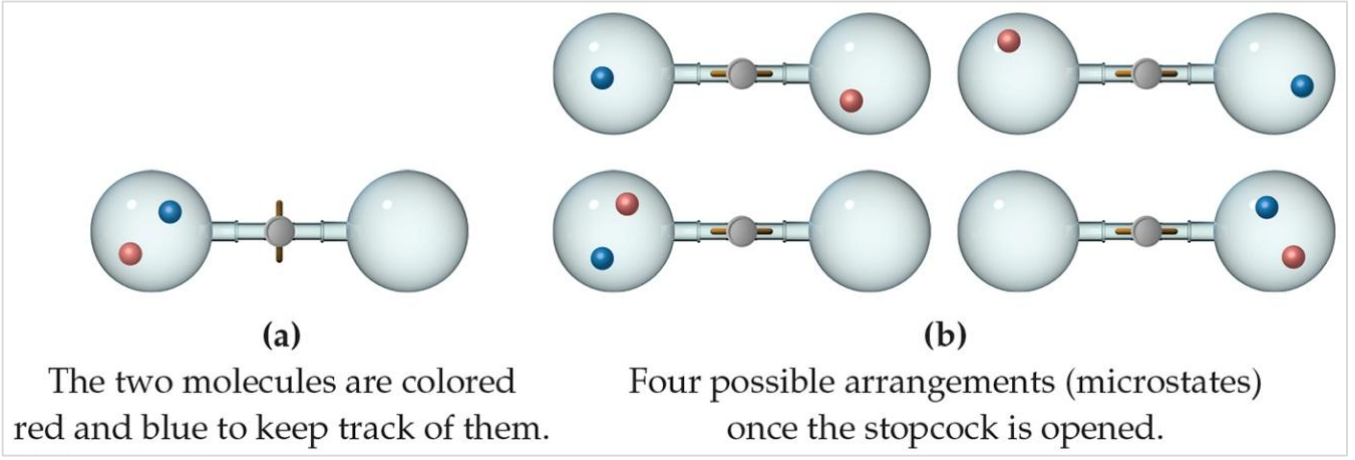
Q. Is ΔS_{sys} positive or negative when steam condenses to liquid water at 100 °C?

Q. For a process, ΔS_{sys} = +42 J/K and ΔS_{surr} = –42 J/K. What can you conclude?

Expansion of a Gas at the Molecular Level

Q. When the stopcock is opened, gas spreads into both flasks. Will all the molecules ever return to flask A?

- **Initial State(Confined:** Before opening stopcock: all molecules are restricted entirely to the left flask.
- **After Opening Stopcock:** Molecules move randomly throughout the entire apparatus, expanding spontaneously.



(a) The two molecules are colored red and blue to keep track of them.

(b) Four possible arrangements (microstates) once the stopcock is opened.

The probability of finding all in the same flask:

- ✓ 2 Molecules: $(½)^2 = 25\%$
- ✓ 3 Molecules: $(½)^3 = 12.5\%$
- ✓ 1 Mole (6.02×10^{23}): $(½)^N \approx 0$

Driving Force: Disorder:

- Spreading increases disorder. Any given molecule gains freedom across both flasks, making the expanded state overwhelmingly more probable.
- Spontaneous expansion is driven by probability

Boltzmann's Equation and Microstates

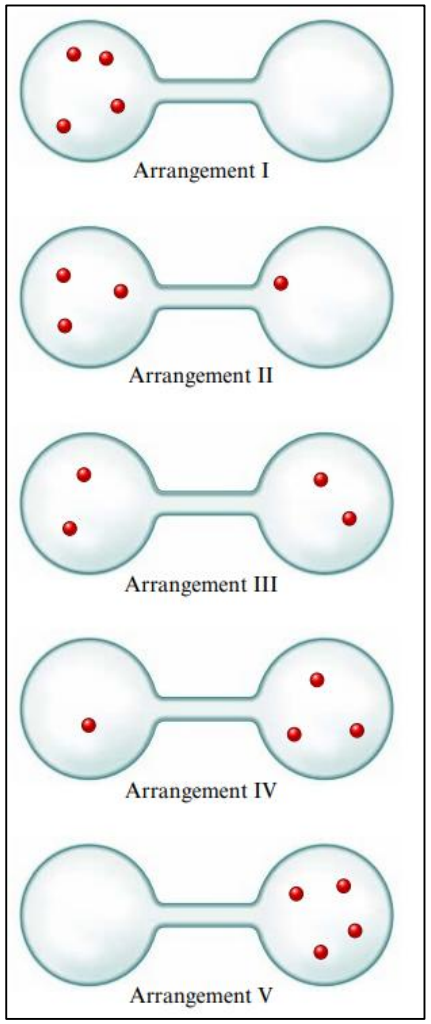


TABLE 17.1 | The Microstates That Give a Particular Arrangement (State)

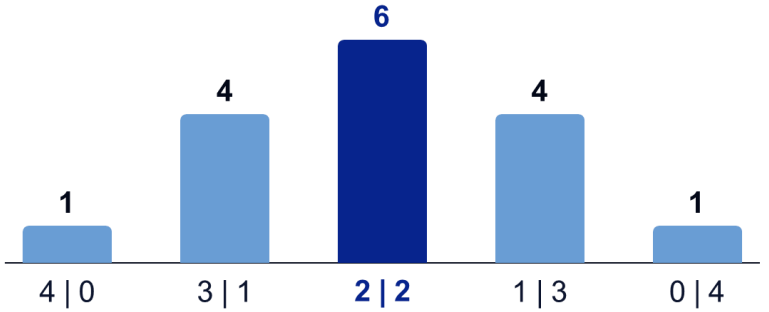
Arrangement	Microstates	Number of Microstates
I		1
II		4
III		6
IV		4
V		1

- **Microstate (W):** one specific arrangement of the particles.
- **Macrostate:** arrangements that look alike

$$W(\text{total}) = n^N$$

N particles, n boxes: $2^4 = 16$

Q. Which arrangement is most likely to occur?



The even split, 2 | 2, is most probable: 6 of 16 microstates.

(Source: Chemistry: Steven S. Zumdahl, Susan A. Zumdahl, and Donald J. DeCoste)

Boltzmann's Equation and Microstates

- Boltzmann imagined a snapshot of the positions and motions of all molecules at one instant; he called this a microstate
- Each thermodynamic state has a specific number of microstates, W
- **Entropy is a measure of how many microstates** are associated with a particular macroscopic state.
- Entropy increases with the number of microstates

$$S = k \ln W$$

$$\Delta S = k \ln W_{\text{final}} - k \ln W_{\text{initial}}$$

$$\Delta S = k \ln \frac{W_{\text{final}}}{W_{\text{initial}}}$$

W : Number of microstates

k : Boltzmann constant (1.38×10^{-23} J/K)

- More microstates in the final state ($W_{final} > W_{initial}$) $\rightarrow$ **$\Delta S > 0$.**
- Entropy increases with the number of microstates!

Boltzmann's Equation and Microstates: Microstates and ΔS

Q. (a) Two gas molecules start in flask *A*; the stopcock is opened. Calculate ΔS.
(b) Calculate ΔS when 1.00 mol of an ideal gas doubles its volume at constant T.
($k = 1.38 \times 10^{-23}$ J/K; $R = N_A k = 8.314$ J/mol·K)

(a) $W_{initial} = 1$ (both in *A*); $W_{final} = 4$
$\Delta S = k \ln 4 = (1.38 \times 10^{-23})(1.386) = \mathbf{1.91 \times 10^{-23}\ J/K}$

(b) Each molecule has twice as many positions: $W_{final}/W_{initial} = 2^N$, where $N = N_A$

$\Delta S = k \ln 2^N = N_A k \ln 2 = R \ln 2 = (8.314\ \text{J/mol·K})(0.693) = \mathbf{+5.76\ J/K}$

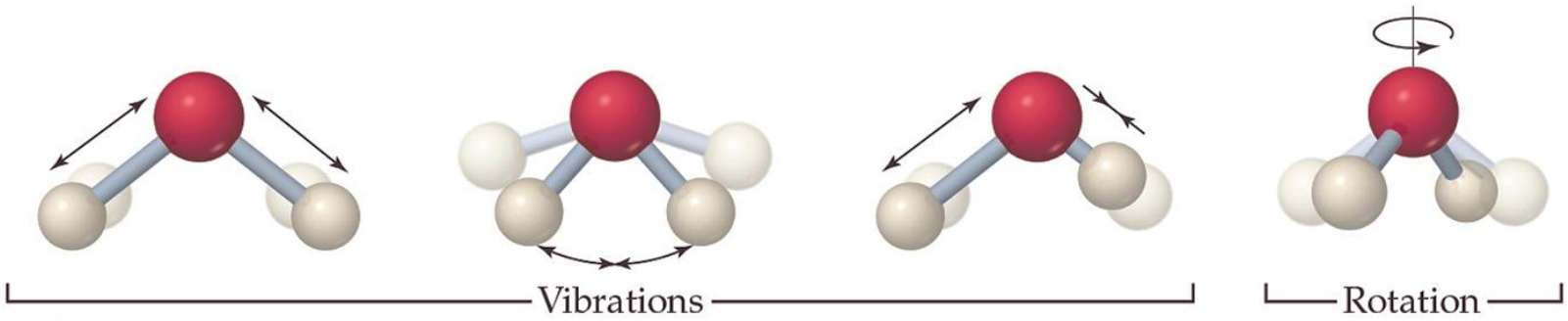
ΔS > 0: expansion increases the number of microstates.

Notice that N_A times k gives the gas constant R; this links the molecular and the macroscopic scales. Expansion increases the number of microstates, so ΔS is positive.

Where do microstates come from? Molecular Motions and Energy

- Molecules store energy in motion:

Translational	**Vibrational**	**Rotational**
Movement of the whole molecule from one place to another	Periodic motion of atoms within a molecule (stretching and bending bonds)	Rotation of the molecule about an axis, or rotation about σ bonds



- More atoms → more possible motions → **more microstates**
- More available modes of motion means more ways to store energy, and more entropy

Factors that raise entropy

Factor	Trend	More microstates from…
▪ Physical state	S(gas) > S(liquid) > S(solid)	more positions, freer motion
▪ Temperature	S rises as T rises	a wider spread of kinetic energies
▪ Atomic mass	heavier atoms → higher S	closer-spaced energy levels
▪ Molecular size	more atoms → higher S	more ways to vibrate
▪ Mixing	mixture > pure parts	more arrangements and interactions

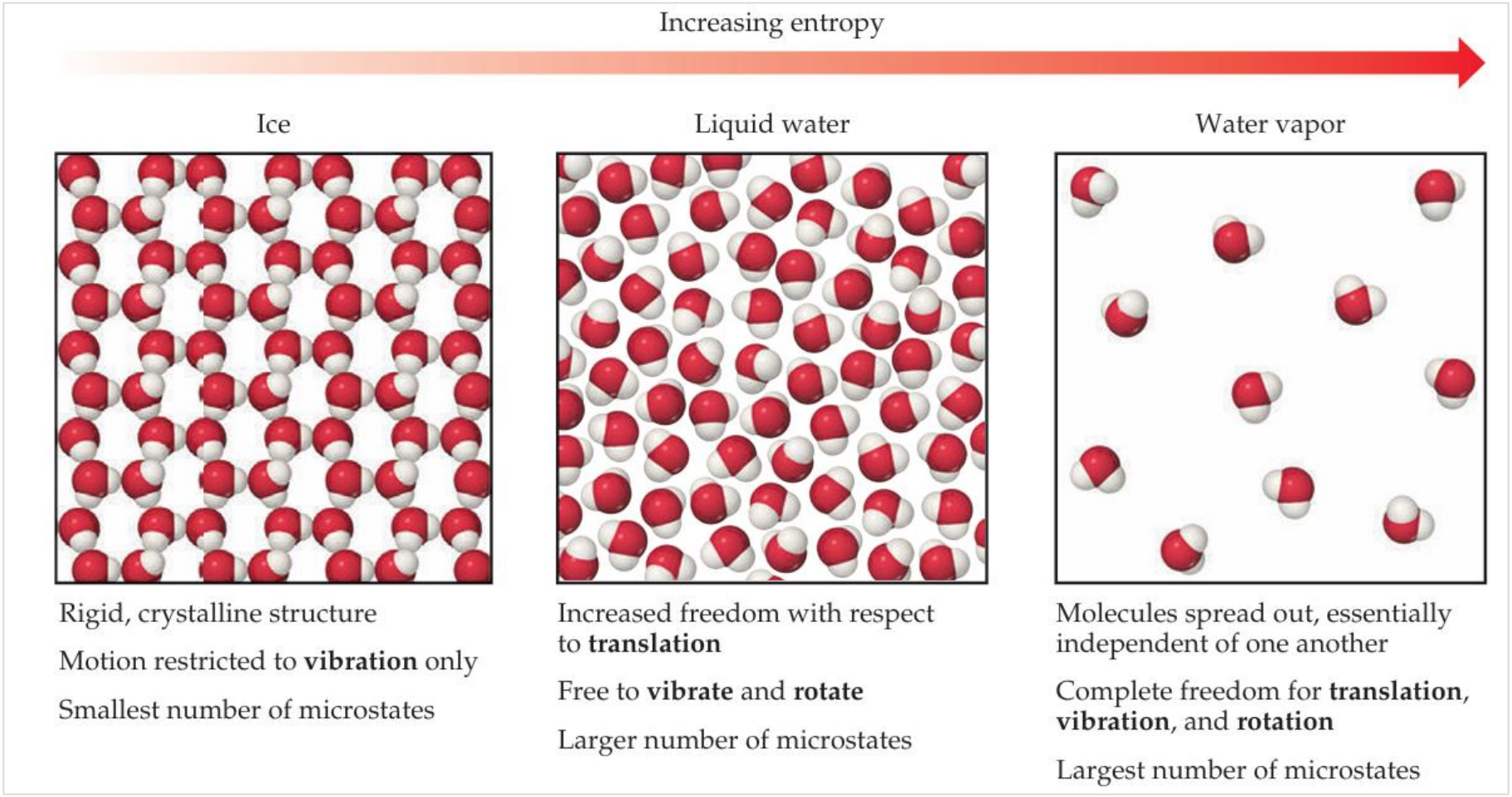
State vs. mass

- At room T, S rises $I_2 \rightarrow Br_2 \rightarrow Cl_2$ although mass falls.
- I_2 is a solid, Br_2 a liquid, Cl_2 a gas.

Rule: S increases with gas formation, with the amount of substance, and with molecular complexity.

Making Qualitative Predictions about ΔS

Q. Which has the highest entropy: ice, liquid water or water vapour?



Entropy and the phases of water. The larger the number of possible microstates, the higher the entropy of the system.

Making Qualitative Predictions about ΔS

Q. Predict whether ΔS is positive or negative for each process at constant temperature:

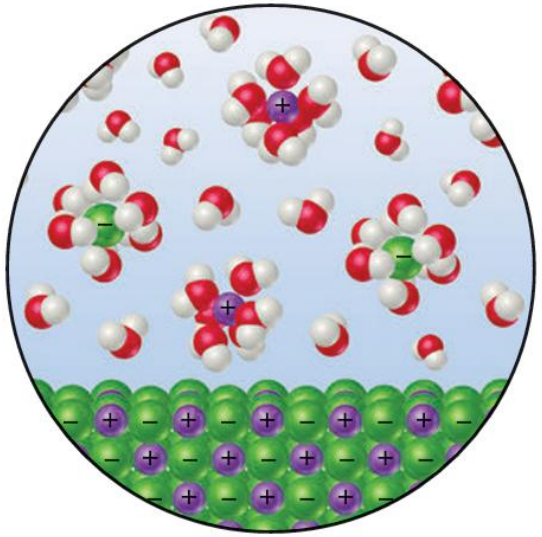
(a) $H_2O(l) \rightarrow H_2O(g)$

(b) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

(c) $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$

(d) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

Note – dissolving a salt (e.g. KCl): two opposing effects – the ions become more spread out (S increases) but the water molecules that hydrate them become more ordered (S decreases). For most salts, the disordering dominates, so $\Delta S > 0$.



▲ **Figure 19.10 Entropy changes when an ionic solid dissolves in water.** The ions become more spread out and disordered, but the water molecules that hydrate the ions become less disordered.

Making Qualitative Predictions about ΔS

Q. In each pair, choose the system that has greater entropy and explain your choice: (a) 1 mol of NaCl(s) or 1 mol of HCl(g) at 25 °C; (b) 2 mol of HCl(g) or 1 mol of HCl(g) at 25 °C; (c) 1 mol of HCl(g) or 1 mol of Ar(g) at 298 K.

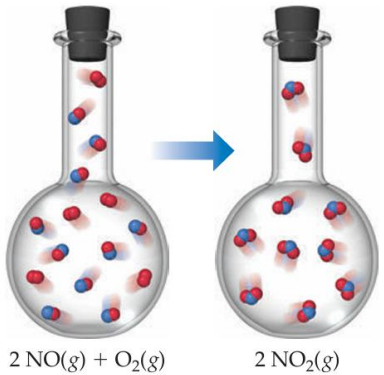
(a) **HCl(g)**: particles in a gas are more disordered and have far more freedom of motion than those in a solid.

(b) **2 mol of HCl(g)**: twice as many molecules – twice the number of microstates and twice the entropy.

(c) **HCl(g)**: a molecule can rotate and vibrate, an Ar atom cannot – more ways to store energy.

Rule: S increases with gas formation, with the amount of substance and with molecular complexity.

Q. What major factor leads to a decrease in entropy as this reaction (when NO(g) is oxidized by $O_2(g)$ to $NO_2(g)$) takes place?



The Third Law of Thermodynamics

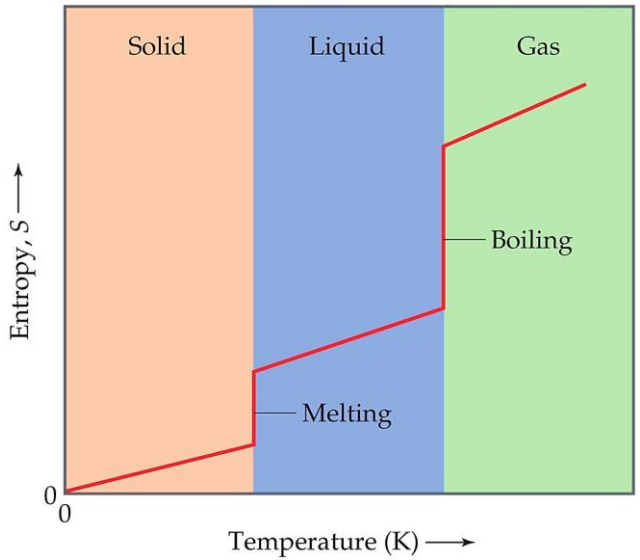
Q. Is there a temperature at which entropy is zero?

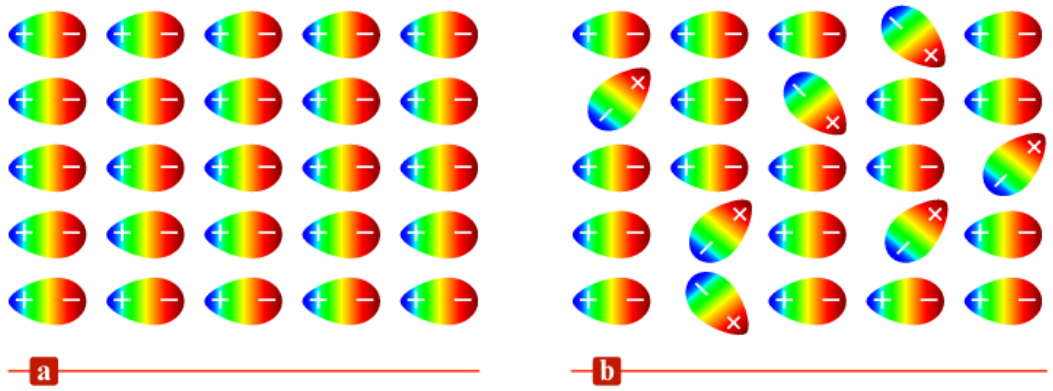
- **Third law:** the entropy of a pure, perfect crystalline substance at absolute zero (0 K) is **zero**.

  Consider phase change in water:

  ✓ At 0 K there is only **one** microstate (W = 1): **S = k ln 1 = 0**

  ✓ As T increases, S rises gradually, then jumps at melting and at boiling (the largest jump).





Hydrogen chloride at 0 K (a) and at temperature above 0 K (b)

Chemistry and Life: Entropy and Human Society

Q. Living things are highly ordered. Does life violate the second law of thermodynamics?

- An organism has far lower entropy than the CO_2, H_2O and simple chemicals it would decompose into.
- **No violation**: the thousands of reactions that build and maintain life increase the entropy of the rest of the universe even more, so over a lifetime $\Delta S_{univ} > 0$.
- Humans also create order around us (technology, nanoscale materials) by spending large amounts of energy.
- Burning petroleum, coal and natural gas to supply this energy releases $CO_2(g)$, $H_2O(g)$ and heat: for every bit of order we produce, we produce an even greater amount of disorder.
- We are using up our store of energy-rich materials – new sources such as solar energy are needed.

19.3 The Molecular Interpretation of Entropy and the Third Law of Thermodynamics

Q. Which has the larger standard entropy: $CH_4(g)$ or $C_2H_6(g)$? Why?

Answer: **$C_2H_6(g)$**: more atoms, so more vibrational and rotational motions (more microstates).

Q. A refrigerator freezes water into ice ($\Delta S_{sys} < 0$). How is the second law still obeyed?

Answer: It releases more heat into the kitchen than it removes from the water, so ΔS_{surr} is larger and positive: $\Delta S_{univ} > 0$.

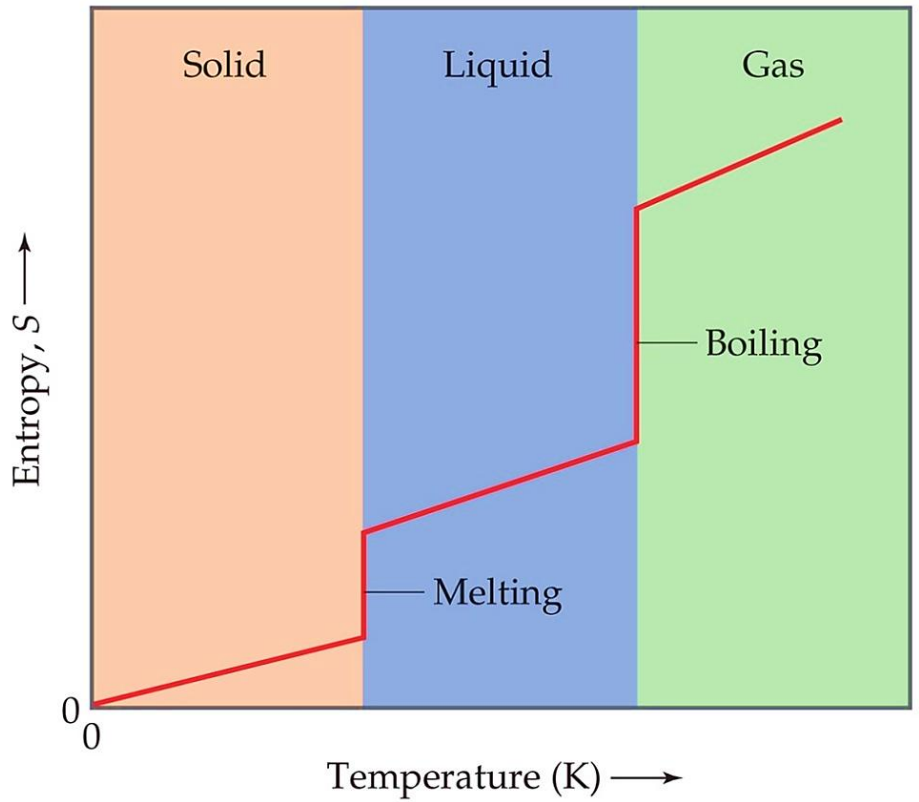
Q. Unlike $\Delta H°_f$, the standard entropy of an element such as $O_2(g)$ is NOT zero at 25 °C. Why?

Answer: Entropy has an absolute zero point (third law: S = 0 only for a perfect crystal at 0 K). At 298 K every substance has $S° > 0$.

19.4 Entropy Changes in Chemical Reactions

Temperature Variation of Entropy

Q. How does the entropy of a substance change as it is heated from 0 K?



Solid

S rises steadily as vibrations about lattice sites increase

Liquid

S rises as T increases

Gas

S keeps rising: more energy in translational motion

Melting

Sharp jump: molecules become free to move through the whole volume

Boiling

Larger jump: far more volume available in the gas

Q. Why is the entropy jump at the boiling point larger than at the melting point?

19.4 Entropy Changes in Chemical Reactions

Standard Molar Entropies (S°)

- **S°:** Molar entropy of one mole of a substance in its standard state (the pure substance at 1 atm)

Q. What patterns can you see in Table 19.1?

- ✓ S° of elements is **not** zero at 298 K (reference is 0 K).
- ✓ S° of gases > liquids > solids: $H_2O(g)$ vs $H_2O(l)$.
- ✓ S° increases with molar mass: Li < Na < K.
- ✓ S° increases with the number of atoms in the formula (more motions).

Rule: S increases with gas formation, with the amount of substance, and with molecular complexity.

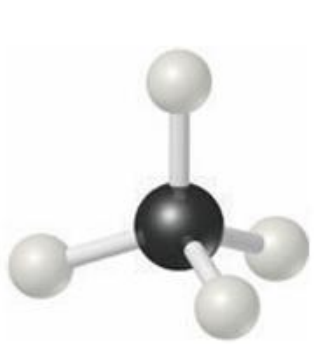
TABLE 19.1 Standard Molar Entropies of Selected Substances at 298 K

Substance	$S°$ (J/mol K)
$H_2(g)$	130.6
$N_2(g)$	191.5
$O_2(g)$	205.0
$H_2O(g)$	188.8
$NH_3(g)$	192.5
$CH_3OH(g)$	237.6
$C_6H_6(g)$	269.2
$H_2O(l)$	69.9
$CH_3OH(l)$	126.8
$C_6H_6(l)$	172.8
$Li(s)$	29.1
$Na(s)$	51.4
$K(s)$	64.7
$Fe(s)$	27.23
$FeCl_3(s)$	142.3
$NaCl(s)$	72.3

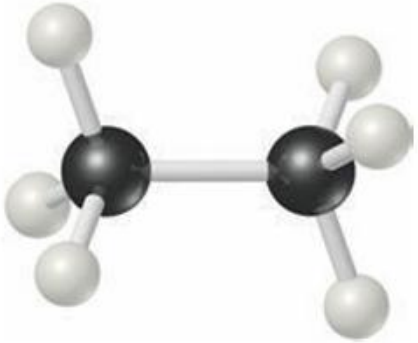
Appendix C gives a more extensive list.

19.4 Entropy Changes in Chemical Reactions

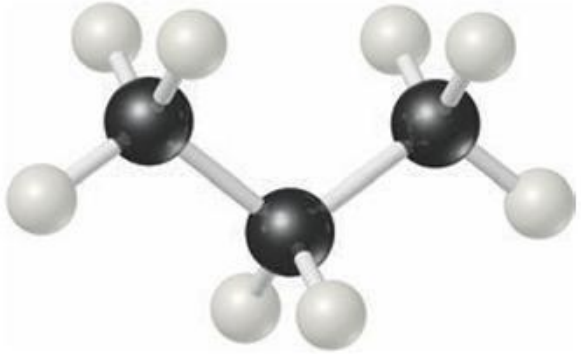
Standard Molar Entropies (S°)



Methane, CH_4
$S° = 186.3$ J/mol K



Ethane, C_2H_6
$S° = 229.6$ J/mol K



Propane, C_3H_8
$S° = 270.3$ J/mol K

Entropy increases with increasing molecular complexity.

- S° generally increases with the **number of atoms** in the formula: more atoms → more vibrational and rotational motions → more microstates.

Q. What might you expect for the value of S° for butane, C_4H_{10}?

Answer: larger than 270.3 J/mol·K – butane has more atoms than propane.

19.4 Entropy Changes in Chemical Reactions

Calculating the Standard Entropy Change for a Reaction

Q. How do we calculate the entropy change for a reaction?

- Apply the same approach as $\Delta H°$ from $\Delta H°_f$ (Chapter 5), but use $S°$ values, **not** $\Delta S°_f$.

- The entropy change in a chemical reaction equals the sum of the entropies of the products **minus** the sum of the entropies of the reactants:

$$\Delta S° = \Sigma\, nS°(\text{products}) - \Sigma\, mS°(\text{reactants})$$

- ✓ n and m are the coefficients of the balanced equation.
- ✓ Elements have $S° \neq 0$: do not leave them out!
- ✓ Units: J/K (per mole of reaction)
- ✓ Remember to convert to kJ when combining with ΔH.

19.4 Entropy Changes in Chemical Reactions

Calculating the Standard Entropy Change for a Reaction

Q. Calculate $\Delta S°$ for the synthesis of ammonia at 298 K: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

($S°$(J/mol·K): $N_2(g)$ 191.5; $H_2(g)$ 130.6; $NH_3(g)$ 192.5)

$$\Delta S° = \Sigma\, nS°(\text{products}) - \Sigma\, mS°(\text{reactants})$$

Step 1. Products: $2S°(NH_3) = 2(192.5) = 385.0$ J/K

Step 2. Reactants: $S°(N_2) + 3S°(H_2) = 191.5 + 3(130.6) = 583.3$ J/K

Step 3. $\Delta S° = 385.0 - 583.3 =$ **-198.3 J/K**

Step 4. Negative, as expected: 4 mol of gas → 2 mol of gas.

19.4 Entropy Changes in Chemical Reactions

Entropy Changes in the Surroundings

Q. How does a reaction change the entropy of its surroundings?

- Heat that flows into or out of the system changes the entropy of the surroundings.
  - ✓ Heat released by the system is absorbed by the surroundings, and vice versa.
  - ✓ For an isothermal process:

$$\Delta S_{surr} = \frac{q_{surr}}{T} = \frac{-q_{sys}}{T}$$

  - ✓ At constant pressure, $q_{sys} = \Delta H_{sys}$, so:

$$\Delta S_{surr} = \frac{-\Delta H_{sys}}{T} \quad [\text{at constant P}]$$

  - ✓ Exothermic reaction ($\Delta H < 0$) → $\Delta S_{surr} > 0$.
  - ✓ Endothermic ($\Delta H > 0$) → $\Delta S_{surr} < 0$.
  - ✓ The effect is larger at low temperature (dividing by a smaller T).

Exothermic reactions increase the entropy of the surroundings; endothermic reactions decrease it.

19.4 Entropy Changes in Chemical Reactions

Entropy Changes in the Surroundings

Q. For $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ at 298 K, $\Delta S°_{sys} = -198.3$ J/K. Using $\Delta H°_f(NH_3) = -46.19$ kJ/mol, calculate ΔS_{surr} and ΔS_{univ}. Is the reaction spontaneous?

Step 1. $\Delta H° = 2(-46.19\text{ kJ}) - [0 + 0] = -92.38$ kJ (exothermic)

Step 2. $\Delta S_{surr} = -\Delta H°/T = -(-92{,}380\text{ J})/298\text{ K} =$ **+310.0 J/K**

Step 3. $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = -198.3 + 310.0 =$ **+111.7 J/K**

Step 4. $\Delta S_{univ} > 0 \rightarrow$ **spontaneous** at 298 K, even though $\Delta S_{sys} < 0$.

Q. A reaction has $\Delta H° = +50$ kJ and $\Delta S° = +100$ J/K. Calculate ΔS_{univ} at 298 K. Is it spontaneous?

Answer: $\Delta S_{surr} = -50{,}000/298 = -167.8$ J/K; $\Delta S_{univ} = -67.8$ J/K $\rightarrow$ **not spontaneous** at 298 K.

19.5 Gibbs Free Energy

Why do we need G?

- Spontaneity depends on **both** ΔH and ΔS
  - ✓ Reactions with large negative **ΔH values** tend to be spontaneous
  - ✓ For a spontaneous reaction, the **entropy of the universe must increase**.
- J. Willard Gibbs proposed a new state function, the Gibbs free energy (G) that combines ΔH and ΔS:

$$\mathbf{G = H - TS}$$

ΔG at constant temperature:

$$\mathbf{\Delta G = \Delta H - T\Delta S}$$

- ✓ T is the absolute temperature (kelvin)
- ✓ ΔH is the enthalpy term; TΔS is the entropy term

19.5 Gibbs Free Energy

Connecting ΔG to the entropy of the universe

1. Start from the second law: $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$
2. Substitute $\Delta S_{surr} = -\Delta H_{sys}/T$ (constant T, P): $\Delta S_{univ} = \Delta S_{sys} - \Delta H_{sys}/T$
3. Multiply by –T: $-T\Delta S_{univ} = \Delta H_{sys} - T\Delta S_{sys}$
4. Define the **Gibbs free energy**, G = H – TS. At constant T: $\mathbf{\Delta G = -T\Delta S_{univ}}$

$$\Delta G = \Delta H - T\Delta S$$

- Since $\Delta G = -T\Delta S_{univ}$: $\Delta S_{univ} > 0 \Leftrightarrow \mathbf{\Delta G < 0}$.

When $\Delta S_{universe}$ is positive, ΔG is negative. ΔG is easier to use because it describes only the system, so we do not have to study the surroundings

19.5 Gibbs Free Energy

The Sign of ΔG

Q. What does the sign of ΔG tell us about a reaction at constant T and P?

ΔG < 0 (exergonic)

✓ The forward reaction is spontaneous

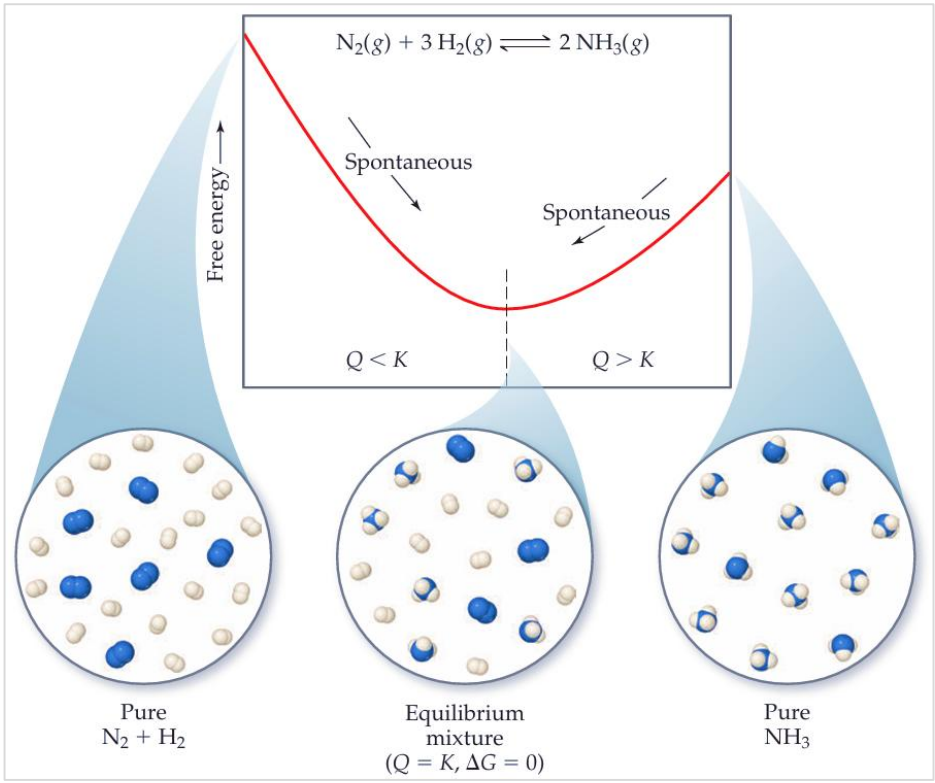
ΔG = 0

✓ The system is at equilibrium

ΔG > 0 (endergonic)

✓ The forward reaction is nonspontaneous

✓ The reverse reaction is spontaneous



- A reaction mixture always moves "downhill" in G toward equilibrium (the minimum of the curve)
  - ✓ Q < K → toward products
  - ✓ Q > K → toward reactants.

19.5 Gibbs Free Energy

The Sign of ΔG: Free Energy Runs Downhill

- A boulder rolls downhill until it reaches the valley – the position of minimum potential energy.
- In the same way, a reacting mixture moves "downhill" in free energy until it reaches the minimum: the **equilibrium mixture**.
- **Free energy always decreases in a spontaneous process** at constant temperature and pressure.
- At the minimum, ΔG = 0: the system is at equilibrium.

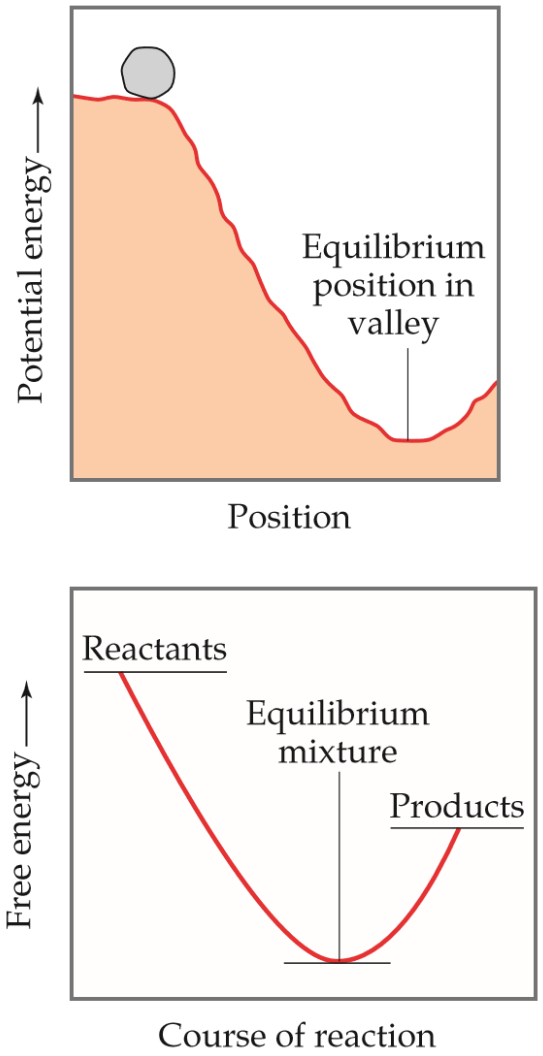


Figure 19.14 Potential energy and free energy.

Q. Are the processes that move a system toward equilibrium spontaneous or nonspontaneous?

Answer: spontaneous – they lower the free energy (from either side).

19.5 Gibbs Free Energy

ΔG° from ΔH°, T and ΔS°

Q. Calculate the standard free-energy change for the formation of NO(g) from $N_2(g)$ and $O_2(g)$ at 298 K: $N_2(g) + O_2(g) \rightarrow 2NO(g)$, given that $\Delta H° = 180.7$ kJ and $\Delta S° = 24.7$ J/K. Is the reaction spontaneous under these conditions?

Step 1. $T\Delta S° = (298\ K)(24.7\ J/K)(1\ kJ/10^3\ J) = 7.4\ kJ$

Step 2. $\Delta G° = \Delta H° - T\Delta S° = 180.7\ kJ - 7.4\ kJ =$ **+173.3 kJ**

Step 3. $\Delta G° > 0$: **not spontaneous** under standard conditions at 298 K.

Note: convert $T\Delta S°$ to kJ before combining it with $\Delta H°$.

19.5 Gibbs Free Energy

ΔG° from ΔH°, T and ΔS°

Q. For $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta H° = -92.38$ kJ and $\Delta S° = -198.3$ J/K. Calculate $\Delta G°$ at 298 K and check it against $\Delta G°_f(NH_3) = -16.66$ kJ/mol.

Step 1. Convert units: $\Delta S° = -198.3$ J/K $= -0.1983$ kJ/K

Step 2. $\Delta G° = \Delta H° - T\Delta S° = -92.38$ kJ $-$ (298 K)(−0.1983 kJ/K) $= -92.38 + 59.09$

Step 3. $\Delta G° =$ **−33.3 kJ** (spontaneous at 298 K)

Step 4. Check: $\Delta G° = 2\Delta G°_f(NH_3) = 2(-16.66) = -33.32$ kJ (agrees)

- In any spontaneous process at constant pressure and temperature, the free energy of the system decreases

19.5 Gibbs Free Energy

Standard Free Energy of Formation (ΔG°_f)

Q. How can we obtain ΔG° for a reaction from tabulated data?

- **ΔG°_f**: ΔG for forming 1 mol of a substance from its elements in their standard states.

$$\Delta G^\circ = \sum n\Delta G_f^\circ(\text{products}) - \sum m\Delta G_f^\circ(\text{reactants})$$

  - ✓ Analogous to standard enthalpies of formation
  - ✓ n and m are the stoichiometric coefficients
  - ✓ ΔG_f° of an element in its standard state is zero

- ΔG° refers to standard conditions (Table 19.2): pure solids and liquids, gases at 1 bar (≈ 1 atm), solutions at 1 M.
- Standard conditions do not include a temperature; data are usually tabulated at 298 K.

TABLE 19.2 Conventions Used in Establishing Standard Free Energies

State of Matter	Standard State
Solid	Pure solid
Liquid	Pure liquid
Gas	100.0 kPa pressure
Solution	1 *M* concentration
Element	$\Delta G_f^\circ = 0$ for element in standard state

19.5 Gibbs Free Energy

Standard Free Energy of Formation

Q. Use $\Delta G°_f$ values to calculate $\Delta G°$ at 298 K for $P_4(g) + 6Cl_2(g) \rightarrow 4PCl_3(g)$. Is the reaction spontaneous under standard conditions? (b) What is $\Delta G°$ for the reverse reaction? ($\Delta G°_f$, kJ/mol: $P_4(g)$ 24.4; $PCl_3(g)$ -269.6)

$$\Delta G° = \sum n\Delta G_f°(\text{products}) - \sum m\Delta G_f°(\text{reactants})$$

Products: $4\Delta G°_f(PCl_3) = 4(-269.6) = -1078.4$ kJ

Reactants: $\Delta G°_f(P_4) + 6\Delta G°_f(Cl_2) = 24.4 + 6(0) = 24.4$ kJ

$\Delta G° = -1078.4 - 24.4 =$ **-1102.8 kJ**

$\Delta G° < 0 \rightarrow$ **spontaneous** under standard conditions.

(b) $4PCl_3(g) \rightarrow P_4(g) + 6Cl_2(g)$: $\Delta G° =$ **+1102.8 kJ**

Rules:

- ✓ Reversing a reaction changes the sign of ΔG° (as for ΔH°).
- ✓ Multiplying the equation by n multiplies ΔG° by n (ΔG° is extensive).

19.5 Gibbs Free Energy

Standard Free Energy of Formation

Q. For $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$, $\Delta H° = -2220$ kJ. (a) Without using data, predict whether $\Delta G°$ is more or less negative than $\Delta H°$. (b) Calculate $\Delta G°$ at 298 K ($\Delta G°_f$, kJ/mol: $CO_2(g)$ –394.4; $H_2O(l)$ –237.13; $C_3H_8(g)$ –23.47).

Step 1. (a) 6 mol of gas → 3 mol of gas (+ liquid): $\Delta S° < 0$, so $-T\Delta S° > 0$ → $\Delta G°$ **less negative** than $\Delta H°$.

Step 2. (b) $\Delta G° = 3\Delta G°_f(CO_2) + 4\Delta G°_f(H_2O, l) - \Delta G°_f(C_3H_8) - 5\Delta G°_f(O_2)$

Step 3. $= 3(-394.4) + 4(-237.13) - (-23.47) - 5(0) =$ **–2108 kJ**

Step 4. –2108 kJ is less negative than –2220 kJ: the prediction is correct. Use $\Delta G°_f$ for $H_2O(l)$, not $H_2O(g)$!

19.5 Gibbs Free Energy

A Closer Look: What's "Free" About Free Energy?

Q. Why is *G* called "free" energy?

Spontaneous process ($\Delta G < 0$)

$-\Delta G$ is the **maximum useful work** the system can do on its surroundings:

$w_{max} = \Delta G$

e.g. the work obtainable from a battery or from burning a fuel

Nonspontaneous process ($\Delta G > 0$)

ΔG is the **minimum work** that must be done on the system to make it happen

e.g. electrolysis of water into H_2 and O_2

- In practice, real (irreversible) processes always deliver **less** than the maximum work; the rest is lost as heat.
- **Example:** the maximum work from burning petrol is given by ΔG, but in a car engine more than 60% of it is lost (mainly as heat); after idling, braking and drag, only about **15%** actually moves the car.

Q. For the combustion of propane, $\Delta G° = -2108$ kJ/mol. What is the maximum useful work obtainable from 1 mol?

Answer: **2108 kJ** (in practice, less).

Q. Use $\Delta G°_f$ values to find $\Delta G°$ for $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$. ($\Delta G°_f\ H_2O(l) = -237.13$ kJ/mol)

Answer: $0 - 2(-237.13) =$ **+474.3 kJ**: nonspontaneous – electrolysis must supply at least 474 kJ of work.

19.6 Free Energy and Temperature

How ΔH and ΔS Determine Spontaneity

Q. For which combinations of ΔH and ΔS does temperature decide the direction of a reaction?

TABLE 19.3 How Signs of ΔH and ΔS Affect Reaction Spontaneity

ΔH	ΔS	$-T\Delta S$	$\Delta G = \Delta H - T\Delta S$	Reaction Characteristics	Example
−	+	−	−	Spontaneous at all temperatures	$2\,O_3(g) \longrightarrow 3\,O_2(g)$
+	−	+	+	Nonspontaneous at all temperatures	$3\,O_2(g) \longrightarrow 2\,O_3(g)$
−	−	+	+ or −	Spontaneous at low T; nonspontaneous at high T	$H_2O(l) \longrightarrow H_2O(s)$
+	+	−	+ or −	Spontaneous at high T; nonspontaneous at low T	$H_2O(s) \longrightarrow H_2O(l)$

ΔG = ΔH – TΔS:

- ✓ The enthalpy term (ΔH) and the entropy term (–TΔS).
- ✓ ΔH < 0, -TΔS < 0: always spontaneous (**ΔG** will always be < 0)
- ✓ ΔH > 0, -TΔS > 0: never spontaneous (**ΔG** will always be > 0)
- ✓ As T increases, the –TΔS term becomes more important.
- ✓ When ΔH and ΔS have the same sign, the reaction changes direction at T = ΔH/ΔS (where ΔG = 0).

19.6 Free Energy and Temperature

ΔG at a Different Temperature

Q. For the Haber process, $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta H° = -92.38$ kJ and $\Delta S° = -198.3$ J/K. Estimate ΔG at 500 °C, assuming $\Delta H°$ and $\Delta S°$ do not change with T. Compare with $\Delta G°$ at 298 K ($\Delta G° = -33.3$ kJ).

Step 1. T = 500 + 273 = 773 K

Step 2. $\Delta G = \Delta H - T\Delta S = -92.38 \text{ kJ} - (773 \text{ K})(-0.1983 \text{ kJ/K}) = -92.38 + 153.3$

Step 3. ΔG = **+60.9 kJ** → **nonspontaneous** at 773 K under standard conditions

$\Delta H < 0$ and $\Delta S < 0$: spontaneous at low T only. It becomes nonspontaneous above $T = 92.38/0.1983 \approx 466$ K.

Q. The Haber process is spontaneous at 25 °C, yet industry runs it at about 500 °C. Why? **Answer:** heating makes it **less** spontaneous ($\Delta S < 0$, so $-T\Delta S$ grows); high T is used for speed (kinetics), not spontaneity.

19.6 Free Energy and Temperature

When Does a Reaction Become Spontaneous?

Q. For $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$, use the data to find $\Delta H°$ and $\Delta S°$, and estimate the temperature above which the decomposition is spontaneous.

($\Delta H°_f$, kJ/mol:- $CaCO_3$: –1207.1, CaO: –635.5, CO_2: –393.5; $S°$, J/mol·K:- $CaCO_3$: 92.88, CaO: 39.75, CO_2 : 213.6)

Step 1. $\Delta H° = (-635.5 - 393.5) - (-1207.1) = +178.1$ kJ (endothermic)

Step 2. $\Delta S° = (39.75 + 213.6) - 92.88 = +160.5$ J/K = 0.1605 kJ/K

Step 3. $\Delta G = 0$ when $T = \Delta H°/\Delta S° = 178.1$ kJ / 0.1605 kJ/K = **1110 K** (≈ 837 °C)

Step 4. $\Delta H > 0$ and $\Delta S > 0$: spontaneous at **high** T – $CaCO_3$ decomposes above ≈ 1110 K (lime kilns).

Q. For a reaction, $\Delta H° = -40.0$ kJ and $\Delta S° = -80.0$ J/K. Below what temperature is it spontaneous?

Answer: T = 40,000/80.0 = **500 K**; spontaneous **below** 500 K ($\Delta H < 0$, $\Delta S < 0$).

19.6 Free Energy and Temperature

Estimating a Normal Boiling Point: Phase changes

- Normal melting and boiling points are temperatures where $\Delta G° = 0$.
- Setting $\Delta G° = \Delta H° - T_b \Delta S° = 0$ gives $\mathbf{T_b = \Delta H°/\Delta S°}$

Q. Use the data to estimate the normal boiling point of $Br_2(l)$: $Br_2(l) \rightarrow Br_2(g)$
($\Delta H°_f$ $Br_2(g)$ = 30.71 kJ/mol; $S°$: $Br_2(l)$ 152.3, $Br_2(g)$ 245.3 J/mol·K)

Step 1. $\Delta H°_{vap}$ = 30.71 – 0 = 30.71 kJ/mol ($Br_2(l)$ is the standard state, $\Delta H°_f = 0$)

Step 2. $\Delta S°_{vap}$ = 245.3 – 152.3 = 93.0 J/mol·K

Step 3. At the normal boiling point, liquid and gas are at equilibrium: $\Delta G° = 0$

Step 4. $T_b = \Delta H°/\Delta S°$ = 30,710 J / 93.0 J/K = **330 K** (57 °C); experimental value: 58.8 °C

19.7 Free Energy and the Equilibrium Constant

Free Energy under Nonstandard Conditions

Q. How does ΔG depend on the composition of the reaction mixture?

$$\Delta G = \Delta G^\circ + RT \ln Q$$

- ✓ R = 8.314 J/mol·K; T in kelvin; Q = reaction quotient (Chapter 15).
- ✓ In Q, use partial pressures (atm) for gases and molar concentrations for solutes; pure solids and liquids are omitted.
- ✓ Standard conditions: all pressures 1 atm, all concentrations 1 M → Q = 1, ln Q = 0 → **$\Delta G = \Delta G^\circ$**.
- ✓ Large Q (much product) makes ΔG more positive; small Q makes ΔG more negative.

Caution:

- Q = 1 gives $\Delta G = \Delta G^\circ$, but this does not mean equilibrium
- At equilibrium $\Delta G = 0$ and Q = K.
- ΔG° is used in kJ/mol, meaning per mole of reaction as written (e.g. per 1 mol N_2, 3 mol H_2, 2 mol NH_3).

19.7 Free Energy and the Equilibrium Constant

Free Energy under Nonstandard Conditions

Q. (a) Write the equation that defines the normal boiling point of $CCl_4(l)$.
(b) What is $\Delta G°$ for this equilibrium?
(c) Estimate the normal boiling point of CCl_4 ($\Delta H°_f$, kJ/mol: $CCl_4(l)$ –139.3, $CCl_4(g)$ –106.7; $S°$, J/mol·K: $CCl_4(l)$ 214.4, $CCl_4(g)$ 309.4).

(a) $CCl_4(l) \rightleftharpoons CCl_4(g)$ at P = 101.3 kPa

(b) At equilibrium $\Delta G = 0$; both phases are in their standard states (Q = 1), so $\boldsymbol{\Delta G° = 0}$.

(c) $\Delta H° = -106.7 - (-139.3) = +32.6$ kJ; $\Delta S° = 309.4 - 214.4 = +95.0$ J/K

$\Delta G° = \Delta H° - T_b\Delta S° = 0 \rightarrow T_b = \Delta H°/\Delta S° = 32{,}600 \text{ J} / 95.0 \text{ J/K} = \boldsymbol{343 \text{ K } (70 °C)}$

Check: Experimental value 76.5 °C – the small difference arises because $\Delta H°$ and $\Delta S°$ change slightly with T.

19.7 Free Energy and the Equilibrium Constant

Free Energy under Nonstandard Conditions

Q. Calculate ΔG at 298 K for the Haber process, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, for a mixture containing 1.0 atm N_2, 3.0 atm H_2 and 0.50 atm NH_3. ($\Delta G° = -33.3$ kJ)

$$\Delta G = \Delta G° + RT \ln Q$$

Step 1. $Q = P_{NH3}^2 / (P_{N2}\, P_{H2}^3) = (0.50)^2 / [(1.0)(3.0)^3] = 9.3 \times 10^{-3}$

Step 2. $RT \ln Q = (8.314 \times 10^{-3}\ \text{kJ/mol·K})(298\ \text{K}) \ln(9.3 \times 10^{-3}) = (2.478)(-4.68) = -11.6$ kJ

Step 3. $\Delta G = -33.3\ \text{kJ} + (-11.6\ \text{kJ}) =$ **-44.9 kJ**

Step 4. ΔG is more negative than $\Delta G°$: with little NH_3 present, the drive to form products is even greater.

19.7 Free Energy and the Equilibrium Constant

Relationship between ΔG° and K

Q. What happens to $\Delta G = \Delta G^\circ + RT \ln Q$ when the system reaches equilibrium?

- At equilibrium: $\Delta G = 0$ and $Q = K$

  $0 = \Delta G^\circ + RT \ln K$

$$\Delta G^\circ = -RT \ln K$$

$$K = e^{-\Delta G^\circ / RT}$$

ΔG°	K	Equilibrium favours
negative	K > 1	products
0	K = 1	neither
positive	K < 1	reactants

- ✓ The more negative ΔG°, the larger K.
- ✓ Use R = 8.314 J/mol·K with ΔG° in J/mol (not kJ!).
- ✓ K is K_p for gases and K_c for solutes (in solution).

19.7 Free Energy and the Equilibrium Constant

Relationship between ΔG° and K

Q. Calculate K for the Haber process $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (a) at 298 K, where $\Delta G° = -33.3$ kJ, and (b) at 773 K, where $\Delta G° \approx +60.9$ kJ.

$$\mathbf{K = e^{-\Delta G°/RT}}$$

(a) $-\Delta G°/RT = 33{,}300\ J / [(8.314\ J/mol{\cdot}K)(298\ K)] = 13.44$

$K = e^{13.44} = \mathbf{6.9 \times 10^5}$ (products strongly favoured)

(b) $-\Delta G°/RT = -60{,}900\ J / [(8.314)(773)] = -9.48$

$K = e^{-9.48} = \mathbf{7.7 \times 10^{-5}}$

- K drops sharply at high T (exothermic reaction) – consistent with Le Châtelier's principle.

Q. For a reaction at 298 K, $K = 1.0 \times 10^3$. Calculate $\Delta G°$.

Answer: $\Delta G° = -(8.314)(298)\ \ln(1.0 \times 10^3) = -(2478)(6.91)\ J = \mathbf{-17.1\ kJ}$

19.7 Free Energy and the Equilibrium Constant

Relationship between ΔG° and K

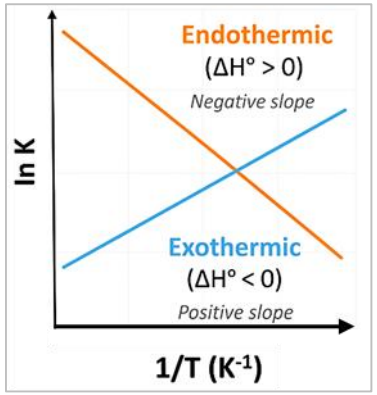
How K shifts with temperature: The Van'T Hoff Equation

$-RT \ln K = \Delta H° - T\Delta S°$, then divide by $-RT$:

$$\ln K = -(\Delta H°/R)(1/T) + \Delta S°/R$$

$$y = m\,x + b$$

Plot	Equals	Solve for
slope	$-\Delta H°/R$	$\Delta H° = -\text{slope} \times R$
intercept	$\Delta S°/R$	$\Delta S° = \text{intercept} \times R$



- ✓ Exothermic: positive slope, **K falls as T rises**.
- ✓ Endothermic: negative slope, **K rises as T rises**.
- ✓ This is Le Châtelier's principle, made quantitative.

Example

A plot of ln K vs 1/T has slope +327.3 K and intercept 5.91.

$\Delta H° = -(327.3)(8.314) =$ **−2.721 kJ/mol**

$\Delta S° = (5.91)(8.314) =$ **+49.1 J/(mol·K)**

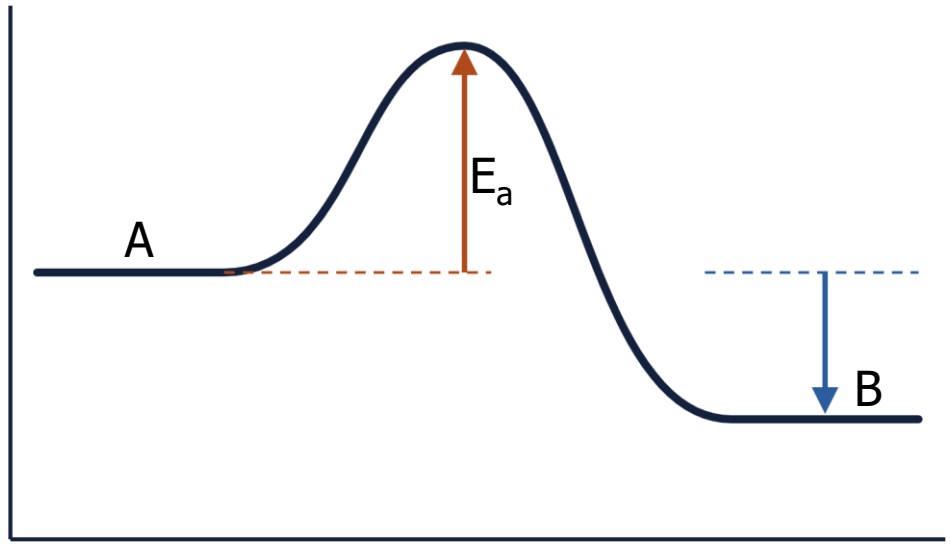
Check: slope −296.1 K, intercept 2.54 → $\Delta H°$ = +2.462 kJ/mol, $\Delta S°$ = +21.1 J/(mol·K)

Assumes $\Delta H°$ and $\Delta S°$ do not vary with T . $\Delta H°$ sets how fast K changes; $\Delta S°$ sets K's size.

19.7 Free Energy and the Equilibrium Constant

Kinetics, Equilibrium, and Stability

- Thermodynamics says "can"; kinetics says "how fast"



E_a, the hill: sets the rate (kinetics)

ΔG°, the drop A → B: sets K (thermodynamics)

Thermodynamic stability

- ✓ ΔG° < 0: reactants are **unstable** relative to products.
- ✓ ΔG° > 0: they are stable.

Kinetic stability

- ✓ High E_a: **inert**, immeasurably slow even if favored.
- ✓ Low E_a: labile, fast.

- A reaction happens on a useful timescale only when the reactants are unstable in **both** senses.
- A catalyst lowers E_a but leaves ΔG° and K unchanged.

19.7 Free Energy and the Equilibrium Constant

Chemistry and Nature: Driving Nonspontaneous Reactions

Q. Copper is extracted from the mineral chalcocite, Cu_2S. The decomposition $Cu_2S(s) \rightarrow 2Cu(s) + S(s)$ has $\Delta G° = +86.2$ kJ. How can copper be obtained spontaneously, using $S(s) + O_2(g) \rightarrow SO_2(g)$, $\Delta G° = -300.4$ kJ?

Step 1. $\Delta G° = +86.2$ kJ > 0: the decomposition is nonspontaneous – Cu cannot be obtained directly.

Step 2. **Couple** (add) it to the spontaneous reaction $S(s) + O_2(g) \rightarrow SO_2(g)$; $S(s)$ cancels.

Step 3. Overall: $Cu_2S(s) + O_2(g) \rightarrow 2Cu(s) + SO_2(g)$ $\Delta G° = (+86.2) + (-300.4) =$ **–214.2 kJ** → spontaneous

Step 4. The spontaneous reaction supplies the free energy that drives the nonspontaneous one – cells use the same principle with ATP (next slide).

19.7 Free Energy and the Equilibrium Constant

Chemistry and Life: Driving Nonspontaneous Reactions (Coupling Reactions)

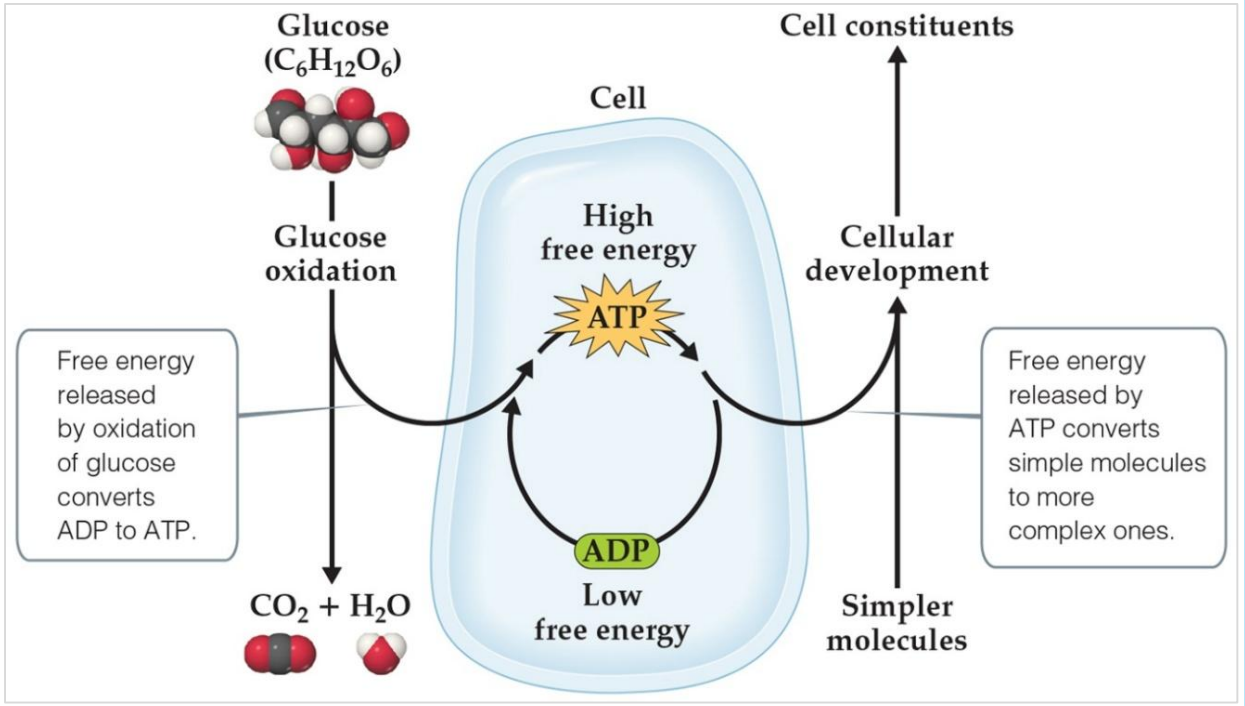
Q. How can cells carry out reactions that are nonspontaneous ($\Delta G > 0$)?

- A nonspontaneous reaction is **coupled** to a strongly spontaneous one, so that the overall $\Delta G < 0$.
- Glucose oxidation releases free energy:

  $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \quad \Delta G° = -2880$ kJ
- It is stored by making ATP from ADP;

  ATP hydrolysis then drives cell reactions:

  $ATP + H_2O \rightarrow ADP + P_i \quad \Delta G° = -30.5$ kJ



Q. A reaction with $\Delta G° = +20$ kJ is coupled to ATP hydrolysis. What is $\Delta G°$ overall?

Answer: +20 + (–30.5) = **–10.5 kJ**: the coupled process is spontaneous.

By now, you should be able to:

- ✓ Distinguish spontaneous from nonspontaneous, and reversible from irreversible, processes.
- ✓ Calculate ΔS for an isothermal process and apply the second law: $\Delta S_{univ} > 0$ for spontaneous change.
- ✓ Explain entropy using microstates ($S = k \ln W$) and predict the sign of ΔS.
- ✓ State the third law and calculate $\Delta S°$ for reactions from standard molar entropies.
- ✓ Calculate $\Delta G°$ from $\Delta G°_f$ values or from $\Delta G° = \Delta H° - T\Delta S°$, and interpret its sign.
- ✓ Predict how temperature affects spontaneity and find the temperature at which $\Delta G = 0$.
- ✓ Use $\Delta G = \Delta G° + RT \ln Q$ and $\Delta G° = -RT \ln K$.

Key relationships

Topic	Key relationship
▪ Entropy (isothermal)	$\Delta S = q_{rev}/T$; phase change: $\Delta S = \Delta H_{trans}/T_{trans}$; gas: $\Delta S = nR \ln(V_2/V_1)$
▪ Second law	$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$ (spontaneous); $= 0$ (equilibrium)
▪ Microstates	$S = k \ln W$; $\Delta S = k \ln(W_{final}/W_{initial})$; $k = 1.38 \times 10^{-23}$ J/K
▪ Third law	$S = 0$ for a perfect crystal at 0 K
▪ Reaction entropy	$\Delta S° = \Sigma\, nS°(\text{products}) - \Sigma\, mS°(\text{reactants})$
▪ Surroundings	$\Delta S_{surr} = -\Delta H_{sys}/T$ (constant P, T)
▪ Gibbs free energy	$\Delta G = \Delta H - T\Delta S$; $\Delta G° = \Sigma\, n\Delta G°_f(\text{products}) - \Sigma\, m\Delta G°_f(\text{reactants})$; $\Delta G = -w_{max}$
▪ Temperature	$\Delta G = 0$ at $T = \Delta H/\Delta S$ (assuming ΔH, ΔS independent of T)
▪ Equilibrium	$\Delta G = \Delta G° + RT \ln Q$; $\Delta G° = -RT \ln K$; $K = e^{-\Delta G°/RT}$

End-of-Chapter Exercises

From the Textbook

Q. Which of the following processes are spontaneous and which are nonspontaneous: (a) mixing of water and ethanol, (b) dissolution of sugar in a cup of hot coffee, (c) formation of oxygen atoms from O_2 molecules at STP, (d) rusting of iron, (e) formation of glucose from CO_2 and H_2O at STP? *[**Exercise 19.57**]*

Q. Indicate whether each statement is true or false. (a) A reaction that is spontaneous in one direction will be nonspontaneous in the reverse direction under the same reaction conditions. (b) All spontaneous processes are fast. (c) Most spontaneous processes are reversible. (d) An isothermal process is one in which the system loses no heat. (e) The maximum amount of work can be accomplished by an irreversible process rather than a reversible one. *[**Exercise 19.58**]*

Q. Consider the melting of ice to liquid water at a pressure of 101.3 kPa. (a) Is this process endothermic or exothermic? (b) In what temperature range is it a spontaneous process? (c) In what temperature range is it nonspontaneous? (d) At what temperature are the two phases in equilibrium? *[**Exercise 19.59**]*

End-of-Chapter Exercises

From the Textbook

Q. The normal boiling point of mercury (Hg) is 356.7 °C, and its molar enthalpy of vaporization is ΔH_{vap} = 59.11 kJ/mol. (a) When Hg boils at its normal boiling point, does its entropy increase or decrease? (b) Calculate the value of ΔS when 2.00 mol of Hg is vaporized at 356.7 °C. *[**Exercise 19.64**]*

Q. (a) What sign for ΔS do you expect when the volume of 0.200 mol of an ideal gas at 27 °C is increased isothermally from an initial volume of 10.0 L? (b) If the final volume is 18.5 L, calculate the entropy change for the process. (c) Do you need to specify the temperature to calculate the entropy change? *[**Exercise 19.66**]*

Q. For each pair, predict which substance has the higher entropy per mole at a given temperature: (a) $I_2(s)$ or $I_2(g)$; (b) $O_2(g)$ at 50.7 kPa or O_2 at 101.3 kPa; (c) 1 mol of N_2 in 22.4 L or 1 mol of N_2 in 44.8 L; (d) $CH_3OH(l)$ or $CH_3OH(s)$. *[**Exercise 19.73**]*

End-of-Chapter Exercises

From the Textbook

Q. Predict the sign of the entropy change of the system for each reaction: (a) $CO(g) + H_2(g) \rightarrow C(s) + H_2O(g)$; (b) $2O_2(g) + N_2(g) \rightarrow 2NO_2(g)$; (c) $NH_4Cl(s) \rightarrow HCl(g) + NH_3(g)$; (d) $2C_2H_4(g) + O_2(g) \rightarrow 2C_2H_4O(g)$. *[**Exercise 19.74**]*

Q. Using S° values from Appendix C, calculate $\Delta S°$ for the following reactions and account for the sign: (a) $NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$; (b) $CH_3OH(g) \rightarrow CO(g) + 2H_2(g)$; (c) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$; (d) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$. *[**Exercise 19.79**]*

Q. For a certain chemical reaction, $\Delta H° = -40.0$ kJ and $\Delta S° = -150.0$ J/K. (a) Does the reaction lead to an increase or decrease in the randomness of the system? (b) Of the surroundings? (c) Calculate $\Delta G°$ for the reaction at 298 K. (d) Is the reaction spontaneous at 298 K under standard conditions? *[**Exercise 19.81**]*

End-of-Chapter Exercises

From the Textbook

Q. Using data from Appendix C, calculate $\Delta G°$ for the following reactions and indicate whether each is spontaneous at 298 K: (a) $2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$; (b) $2NaBr(s) \rightarrow 2Na(g) + Br_2(g)$; (c) $CH_3OH(g) + CH_4(g) \rightarrow C_2H_5OH(g) + H_2(g)$; (d) $2AgNO_3(s) \rightarrow Ag_2O(s) + N_2O(g) + 2O_2(g)$. *[**Exercise 19.83**]*

Q. From the values given for $\Delta H°$ and $\Delta S°$, calculate $\Delta G°$ at 298 K. If a reaction is not spontaneous at 298 K, at what temperature (if any) would it become spontaneous? (a) $2PbS(s) + 3O_2(g) \rightarrow 2PbO(s) + 2SO_2(g)$, $\Delta H° = -844$ kJ, $\Delta S° = -165$ J/K; (b) $2POCl_3(g) \rightarrow 2PCl_3(g) + O_2(g)$, $\Delta H° = 572$ kJ, $\Delta S° = 179$ J/K. *[**Exercise 19.85**]*

Q. Consider the reaction $NO_2(g) + N_2O(g) \rightarrow 3NO(g)$. (a) Use data in Appendix C to predict how ΔG varies with increasing temperature. (b) Calculate ΔG at 800 K, assuming $\Delta H°$ and $\Delta S°$ do not change with temperature. Is the reaction spontaneous? (c) Calculate ΔG at 1000 K. Is it spontaneous at this temperature? *[**Exercise 19.88**]*

End-of-Chapter Exercises

From the Textbook

Q. Consider the reaction $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$. (a) Using data from Appendix C, calculate $\Delta G°$ at 298 K. (b) Calculate ΔG at 298 K if the partial pressures of all gases are 33.4 kPa. *[**Exercise 19.92**]*

Q. Use data from Appendix C to calculate the equilibrium constant, K, and $\Delta G°$ at 298 K for each reaction: (a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; (b) $C_2H_5OH(g) \rightleftharpoons C_2H_4(g) + H_2O(g)$; (c) $3C_2H_2(g) \rightleftharpoons C_6H_6(g)$. *[**Exercise 19.93**]*

Q. The value of K_a for nitrous acid (HNO_2) at 25 °C is 4.5×10^{-4}. (a) Write the chemical equation for the equilibrium that corresponds to K_a. (b) Calculate $\Delta G°$ for the dissociation of nitrous acid. (c) What is ΔG at equilibrium? (d) What is ΔG when $[H^+] = 5.0 \times 10^{-2}$ M, $[NO_2^-] = 6.0 \times 10^{-4}$ M and $[HNO_2] = 0.20$ M? *[**Exercise 19.95**]*